System Cabinet – Contents

MGD Chassis – **M**ulti-**G**enerational **D**ata acquisition chassis – consists of components that generate the timed control signals necessary to control the MR imaging system under the direction of the imaging sequence. (pulse sequence) These control signals drive the gradient output, rf output, as well as gate the rf receive window. Details of the circuit board functionality is described later.

ASC Chassis – **A**mplifier **S**upport **C**ontroller – provides the Narrow and Broadband RF Amplifier interface (Control signals) and RF Power Monitoring. See the ASC help pages for more details.

CAM Chassis – **C**ombined **A**SC and **M**GD Chassis – combines the MGD and ASC functionality into one chassis. See the MGD Chassis or ASC Chassis for the detailed board functionality. The CAM chassis differences to the MGD chassis are described in this help document.

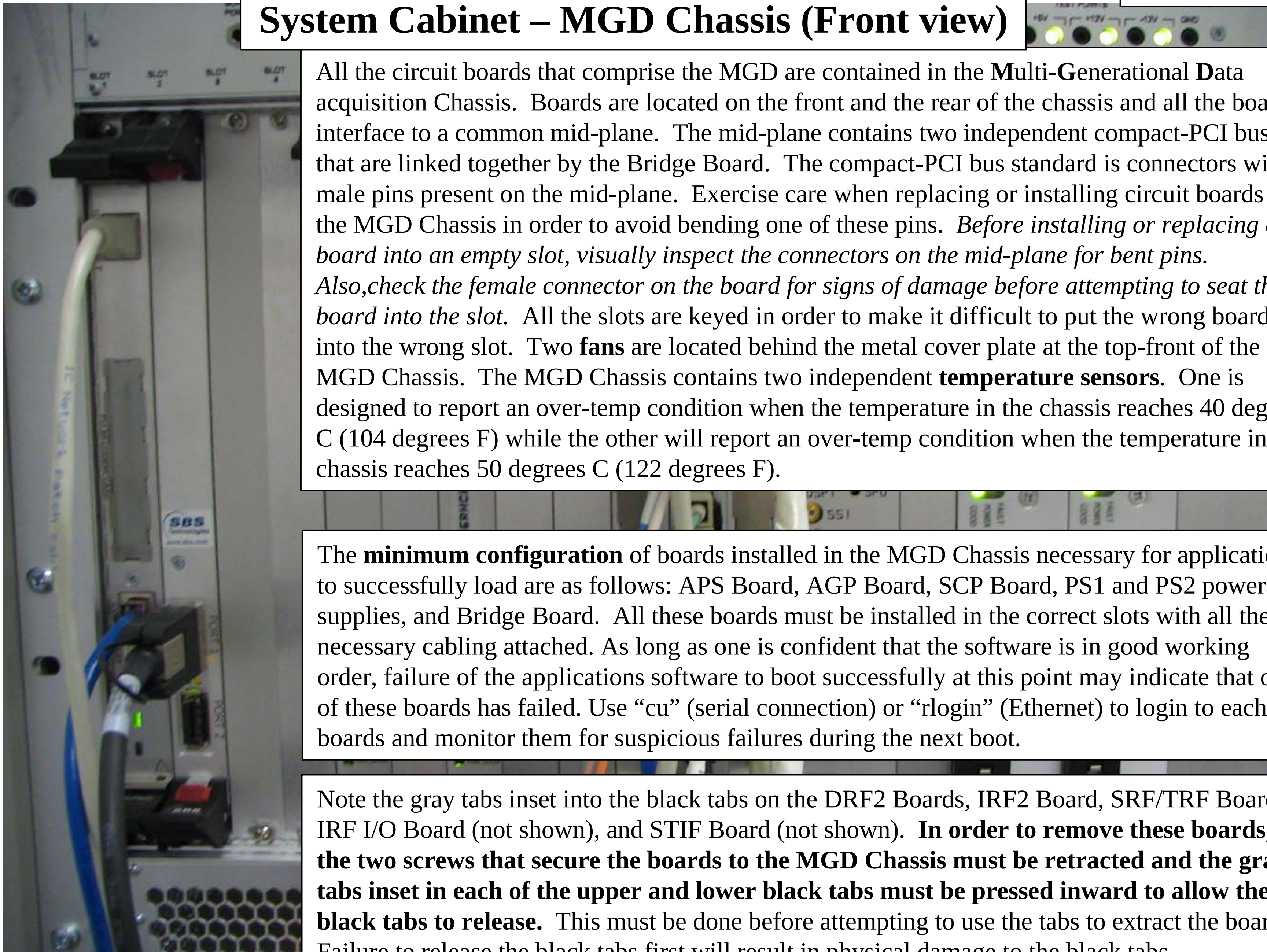
Term Server – permits RS-232C serial connection to each of the processor boards (APS, AGP, SCP) and the STIF Board by utilizing the existing Ethernet link.

VRE – Volume Reconstruction Engine – which reconstructs the image. The VRE is made up of the Infiniband board contained within the MGD or CAM chassis, the Image Compute Nodes (ICNs), and the Network Switch. Some early 4 ICN systems also included an Infiniband Switch. The Infiniband Switch is not necessary anymore and can be removed by rewiring the system.

Network Switch – 16 to 24 port Gigabit Ethernet switch that provides a network path between the Host computer and the ICNs, MGD processors, and the Term Server. This network path is used for application download to the controllers, including the downloading of PSDs to the MGD. It is also used for serial communication between the host and the three (3) MGD processor boards and the STIF board.

ICNs – **I**mage **C**ompute **N**odes – Rack mounted computers that provide the high speed image data to the Host computer.

System Cabinet – MGD Chassis (Front view)



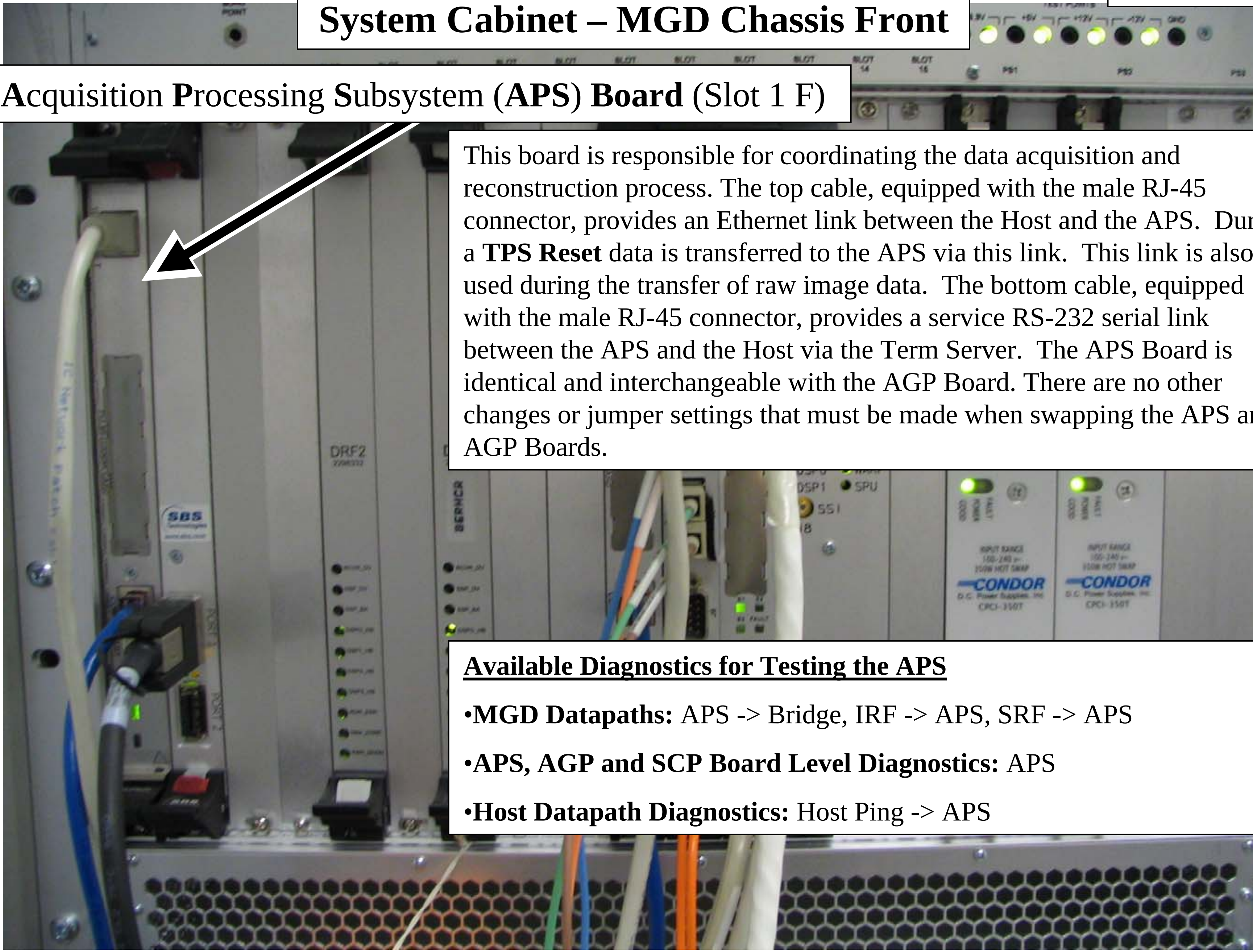
All the circuit boards that comprise the MGD are contained in the **M**ulti-**G**enerational **D**ata acquisition Chassis. Boards are located on the front and the rear of the chassis and all the boards interface to a common mid-plane. The mid-plane contains two independent compact-PCI buses that are linked together by the Bridge Board. The compact-PCI bus standard is connectors with male pins present on the mid-plane. Exercise care when replacing or installing circuit boards in the MGD Chassis in order to avoid bending one of these pins. *Before installing or replacing a board into an empty slot, visually inspect the connectors on the mid-plane for bent pins. Also,check the female connector on the board for signs of damage before attempting to seat the board into the slot.* All the slots are keyed in order to make it difficult to put the wrong board into the wrong slot. Two **fans** are located behind the metal cover plate at the top-front of the MGD Chassis. The MGD Chassis contains two independent **temperature sensors**. One is designed to report an over-temp condition when the temperature in the chassis reaches 40 degrees C (104 degrees F) while the other will report an over-temp condition when the temperature in the chassis reaches 50 degrees C (122 degrees F).

The **minimum configuration** of boards installed in the MGD Chassis necessary for applications to successfully load are as follows: APS Board, AGP Board, SCP Board, PS1 and PS2 power supplies, and Bridge Board. All these boards must be installed in the correct slots with all the necessary cabling attached. As long as one is confident that the software is in good working order, failure of the applications software to boot successfully at this point may indicate that one of these boards has failed. Use “cu” (serial connection) or “rlogin” (Ethernet) to login to each the boards and monitor them for suspicious failures during the next boot.

Note the gray tabs inset into the black tabs on the DRF2 Boards, IRF2 Board, SRF/TRF Board, IRF I/O Board (not shown), and STIF Board (not shown). **In order to remove these boards, the two screws that secure the boards to the MGD Chassis must be retracted and the gray tabs inset in each of the upper and lower black tabs must be pressed inward to allow the black tabs to release.** This must be done before attempting to use the tabs to extract the boards. Failure to release the black tabs first will result in physical damage to the black tabs.

System Cabinet – MGD Chassis Front

Acquisition **P**rocessing **S**ubsystem (**APS**) **Board** (Slot 1 F)



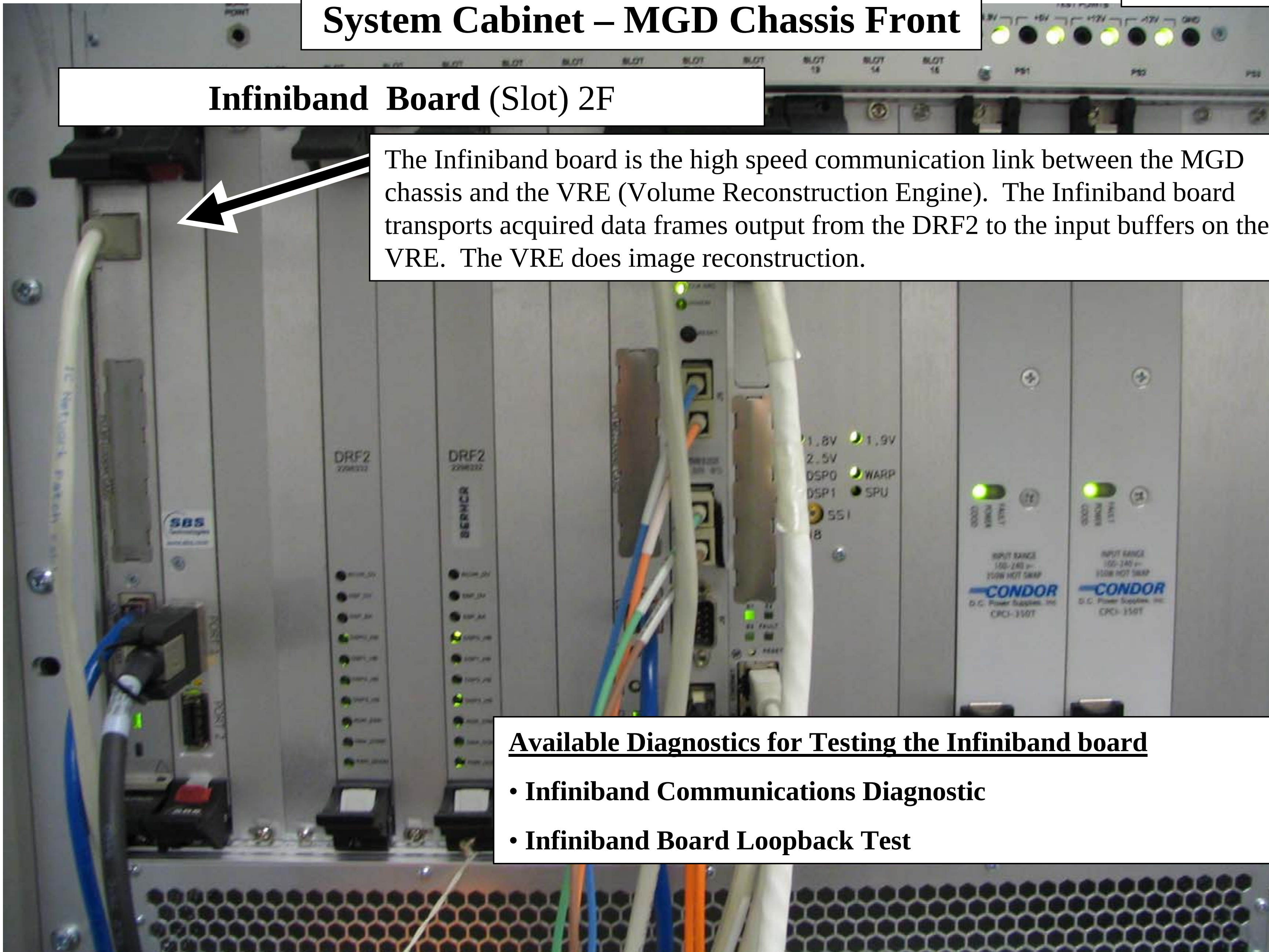
This board is responsible for coordinating the data acquisition and reconstruction process. The top cable, equipped with the male RJ-45 connector, provides an Ethernet link between the Host and the APS. During a **TPS Reset** data is transferred to the APS via this link. This link is also used during the transfer of raw image data. The bottom cable, equipped with the male RJ-45 connector, provides a service RS-232 serial link between the APS and the Host via the Term Server. The APS Board is identical and interchangeable with the AGP Board. There are no other changes or jumper settings that must be made when swapping the APS and AGP Boards.

Available Diagnostics for Testing the APS

- **MGD Datapaths:** APS -> Bridge, IRF -> APS, SRF -> APS
- **APS, AGP and SCP Board Level Diagnostics:** APS
- **Host Datapath Diagnostics:** Host Ping -> APS

System Cabinet – MGD Chassis Front

Infiniband Board (Slot) 2F

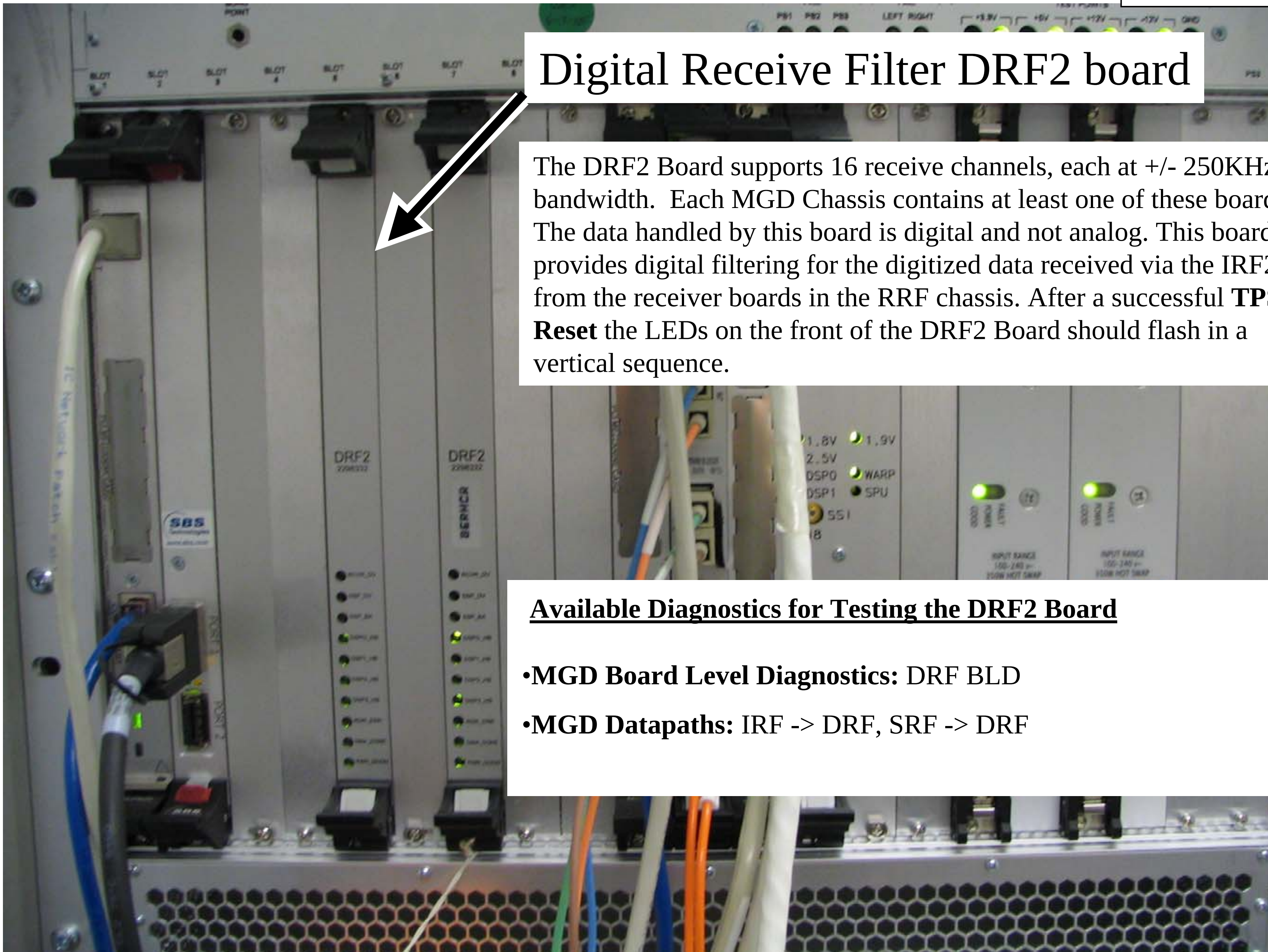


The Infiniband board is the high speed communication link between the MGD chassis and the VRE (Volume Reconstruction Engine). The Infiniband board transports acquired data frames output from the DRF2 to the input buffers on the VRE. The VRE does image reconstruction.

<u>Available Diagnostics for Testing the Infiniband board</u>

- **Infiniband Communications Diagnostic**
- **Infiniband Board Loopback Test**

Digital Receive Filter DRF2 board



The DRF2 Board supports 16 receive channels, each at +/- 250KHz bandwidth. Each MGD Chassis contains at least one of these boards. The data handled by this board is digital and not analog. This board provides digital filtering for the digitized data received via the IRF2 from the receiver boards in the RRF chassis. After a successful **TPS Reset** the LEDs on the front of the DRF2 Board should flash in a vertical sequence.

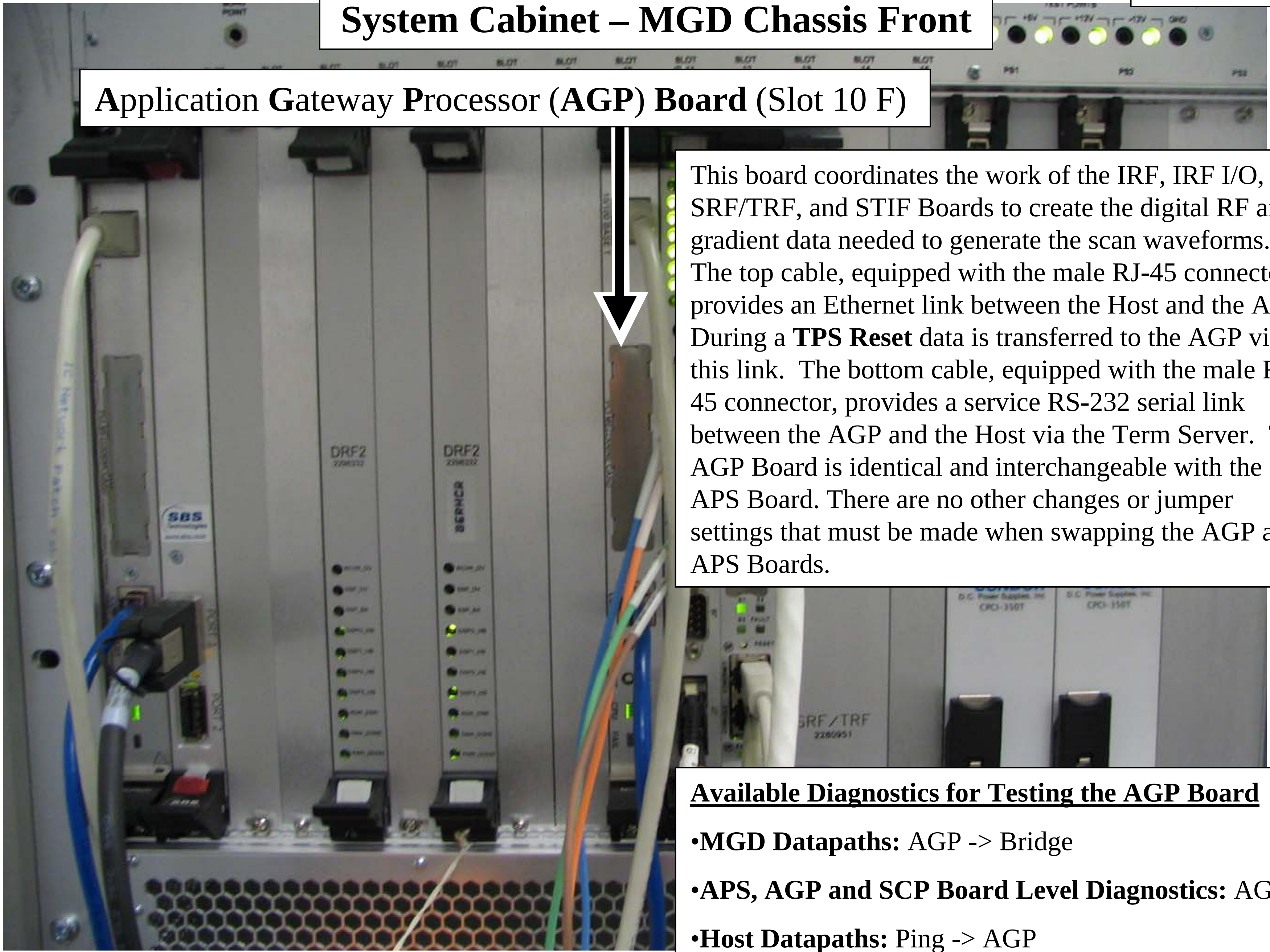
<u>Available Diagnostics for Testing the DRF2 Board</u>

•**MGD Board Level Diagnostics:** DRF BLD

•**MGD Datapaths:** IRF -> DRF, SRF -> DRF

System Cabinet – MGD Chassis Front

Application **G**ateway **P**rocessor (**AGP**) **Board** (Slot 10 F)



This board coordinates the work of the IRF, IRF I/O, SRF/TRF, and STIF Boards to create the digital RF and gradient data needed to generate the scan waveforms. The top cable, equipped with the male RJ-45 connector, provides an Ethernet link between the Host and the AGP. During a **TPS Reset** data is transferred to the AGP via this link. The bottom cable, equipped with the male RJ-45 connector, provides a service RS-232 serial link between the AGP and the Host via the Term Server. The AGP Board is identical and interchangeable with the APS Board. There are no other changes or jumper settings that must be made when swapping the AGP and APS Boards.

<u>Available Diagnostics for Testing the AGP Board</u>

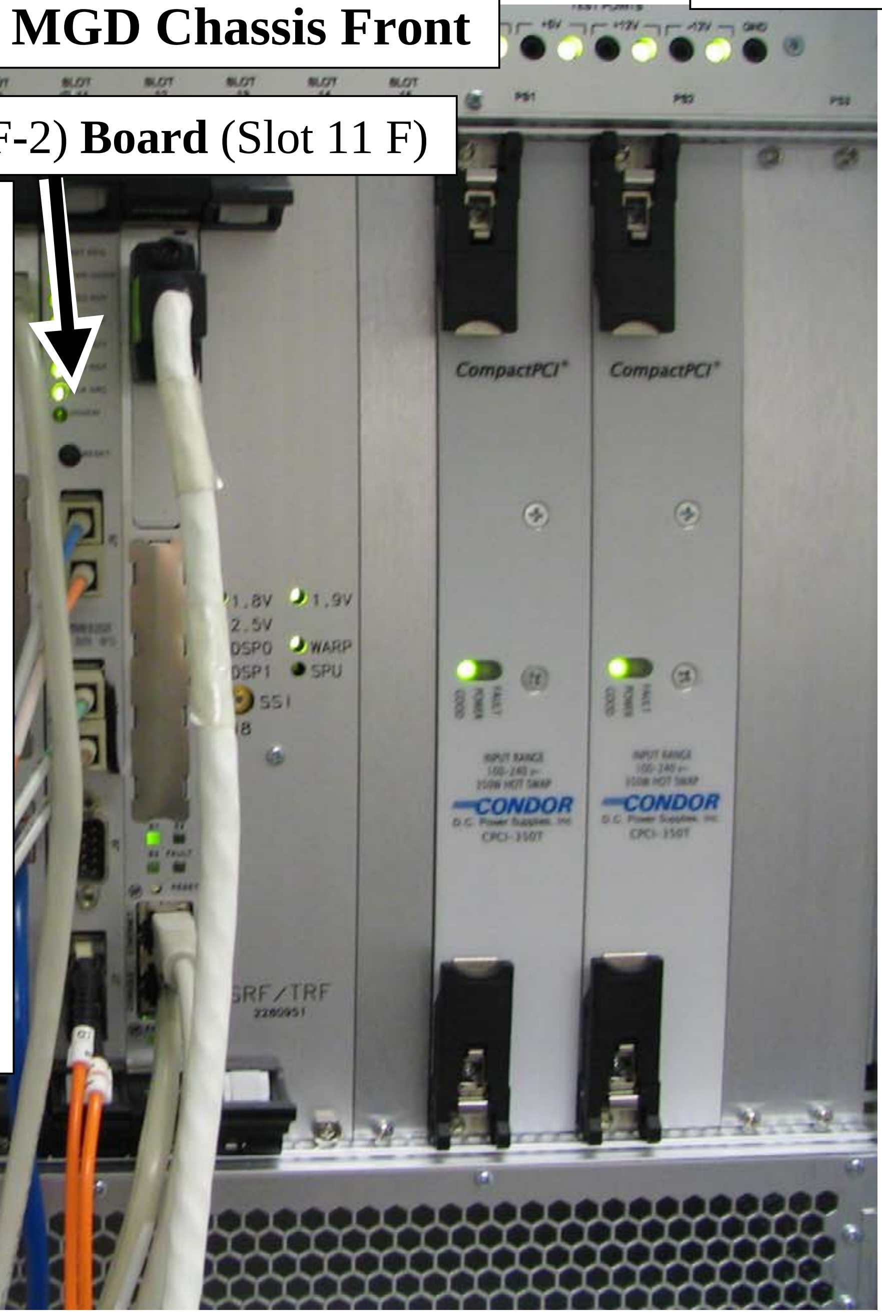
•**MGD Datapaths:** AGP -> Bridge

•**APS, AGP and SCP Board Level Diagnostics:** AGP

•**Host Datapaths:** Ping -> AGP

System Cabinet – MGD Chassis Front

Interface & **R**emote **F**unctions (IRF-2) **Board** (Slot 11 F)



The IRF2 provides a master data acquisition clock signal to the rest of the MGD components. This signal is obtained from a 20MHz reference provided by the RRF Master Exciter. The IRF Board transfers digitized data from the SRF/TRF Board to the Exciter(s) in the RRF so that the RF waveforms can be generated. The IRF2 Board receives view data from the receivers that it sends to the DRF2(s). The IRF Board also provides for control and status monitoring of coil temperature, PDU mode, scan room door status, bore fan, bore lights, and alignment lights. The RESET button on the front of this board can be used to initiate a manual reset of the MGD if the serial reset line is not working. Press this button and then either TPS Reset or boot applications. Timing is critical so if it does not work the first time, try again.

The IRF-2 board is needed for HDx as the interface to the single DRF-2 board as it enables the interface for 16 channel option. Software will not support the IRF-1 board in 14x.

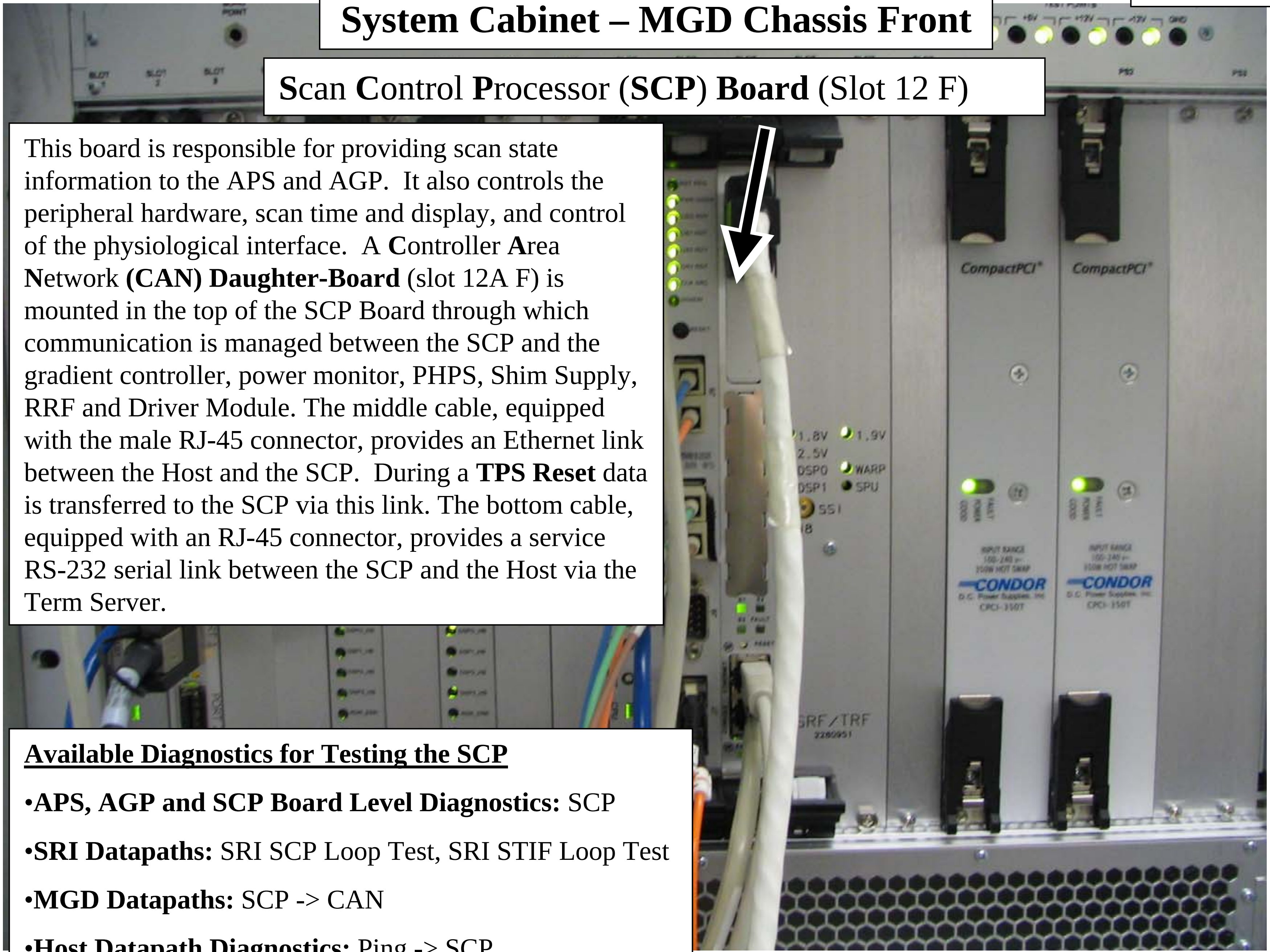
Available Diagnostics for Testing the IRF Board

•**MGD Board Level Diagnostics:** IRF BLD

•**MGD Datapaths:** IRF -> DRF, IRF -> IRF I/O, IRF -> APS, SRF -> IRF

System Cabinet – MGD Chassis Front

Scan **C**ontrol **P**rocessor (**SCP**) **Board** (Slot 12 F)



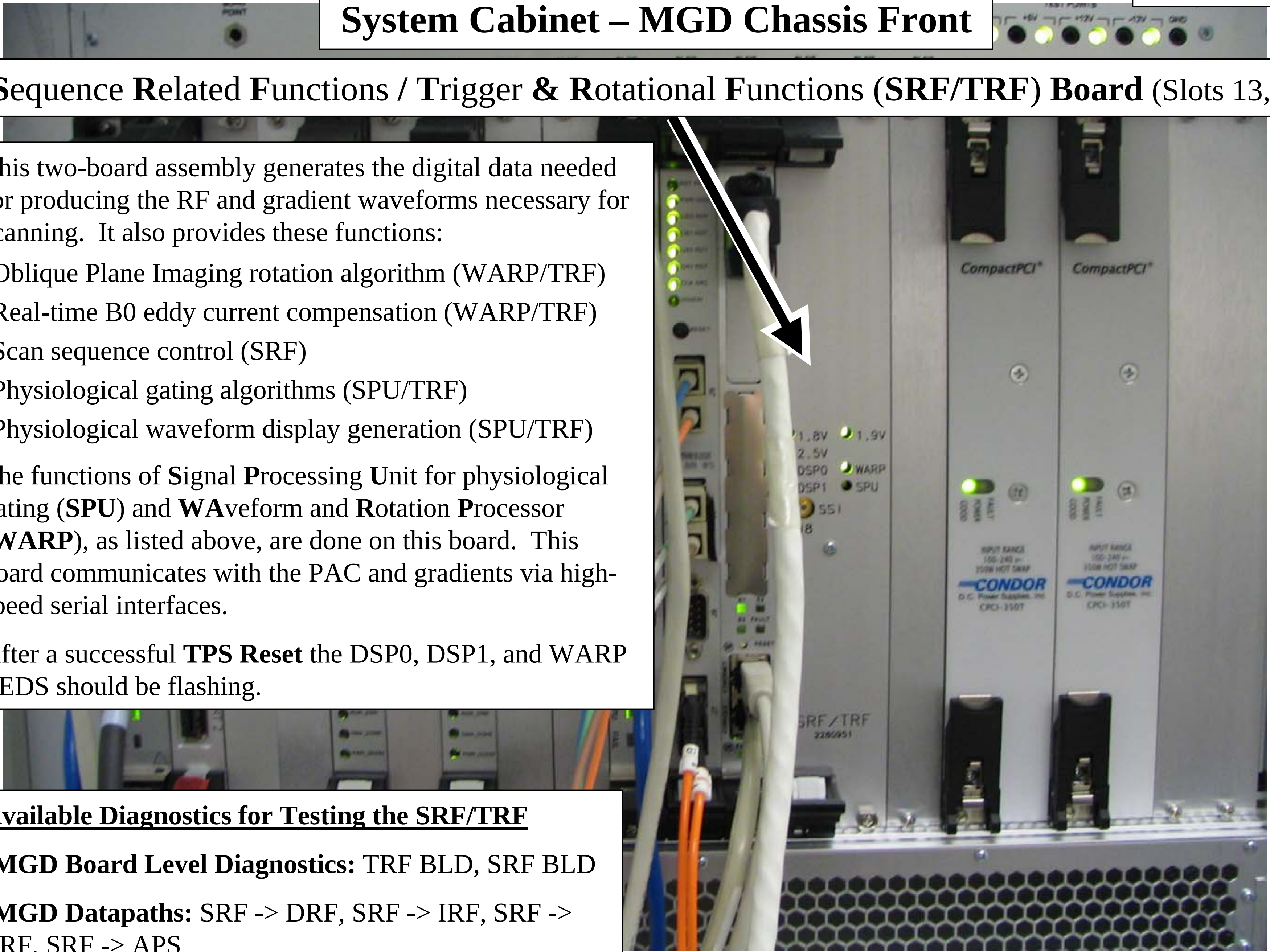
This board is responsible for providing scan state information to the APS and AGP. It also controls the peripheral hardware, scan time and display, and control of the physiological interface. A **C**ontroller **A**rea **N**etwork **(CAN) Daughter-Board** (slot 12A F) is mounted in the top of the SCP Board through which communication is managed between the SCP and the gradient controller, power monitor, PHPS, Shim Supply, RRF and Driver Module. The middle cable, equipped with the male RJ-45 connector, provides an Ethernet link between the Host and the SCP. During a **TPS Reset** data is transferred to the SCP via this link. The bottom cable, equipped with an RJ-45 connector, provides a service RS-232 serial link between the SCP and the Host via the Term Server.

<u>Available Diagnostics for Testing the SCP</u>

- **APS, AGP and SCP Board Level Diagnostics:** SCP
- **SRI Datapaths:** SRI SCP Loop Test, SRI STIF Loop Test
- **MGD Datapaths:** SCP -> CAN
- **Host Datapath Diagnostics:** Ping -> SCP

System Cabinet – MGD Chassis Front

Sequence **R**elated **F**unctions / **T**rigger **&** **R**otational **F**unctions (**SRF/TRF**) **Board** (Slots 13,14 F)



This two-board assembly generates the digital data needed for producing the RF and gradient waveforms necessary for scanning. It also provides these functions:

- •Oblique Plane Imaging rotation algorithm (WARP/TRF)
- •Real-time B0 eddy current compensation (WARP/TRF)
- •Scan sequence control (SRF)
- •Physiological gating algorithms (SPU/TRF)
- •Physiological waveform display generation (SPU/TRF)

The functions of **S**ignal **P**rocessing **U**nit for physiological gating (**SPU**) and **WA**veform and **R**otation **P**rocessor (**WARP**), as listed above, are done on this board. This board communicates with the PAC and gradients via high-speed serial interfaces.

After a successful **TPS Reset** the DSP0, DSP1, and WARP LEDS should be flashing.

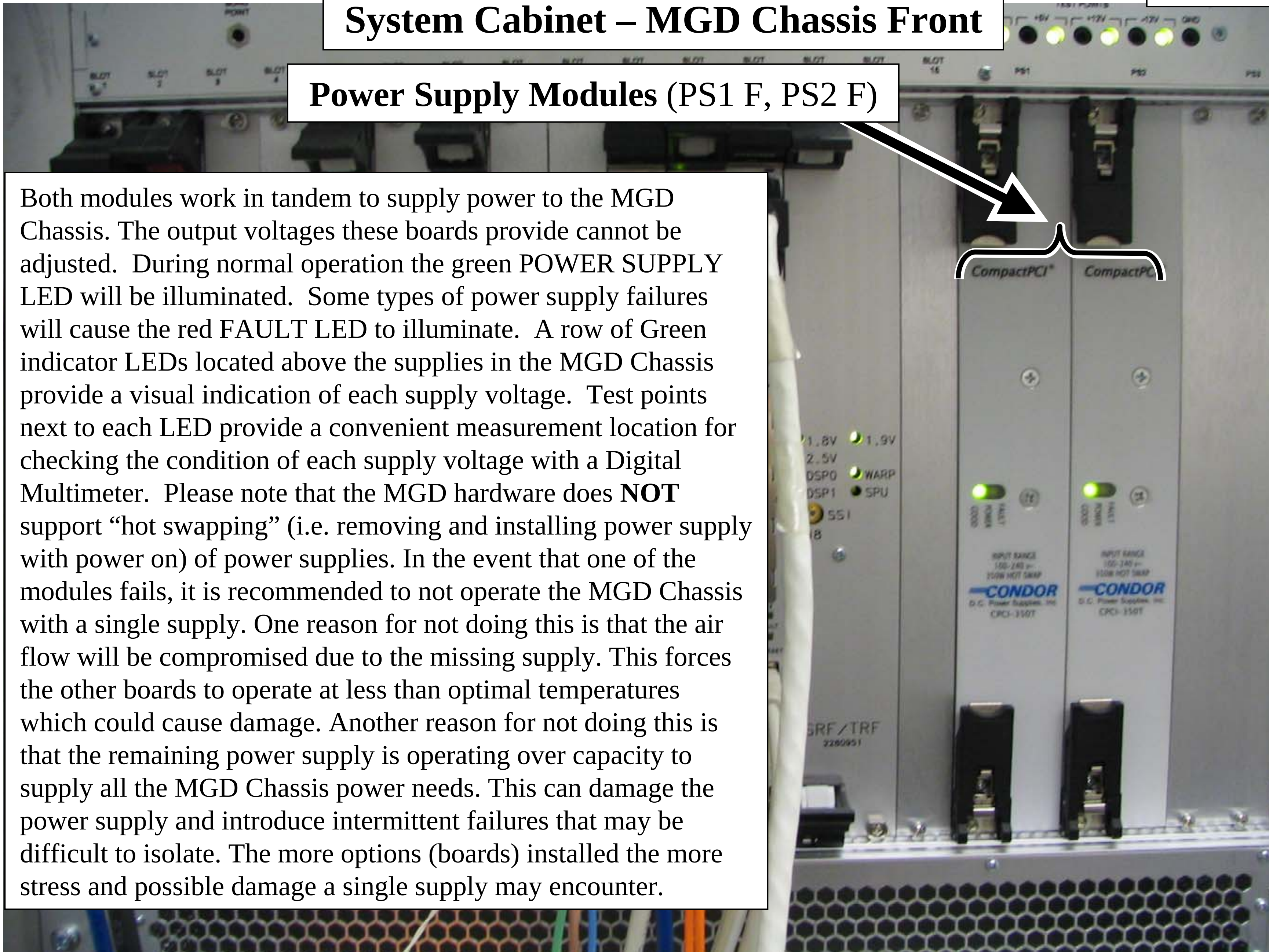
<u>Available Diagnostics for Testing the SRF/TRF</u>

•**MGD Board Level Diagnostics:** TRF BLD, SRF BLD

•**MGD Datapaths:** SRF -> DRF, SRF -> IRF, SRF -> TRF, SRF -> APS

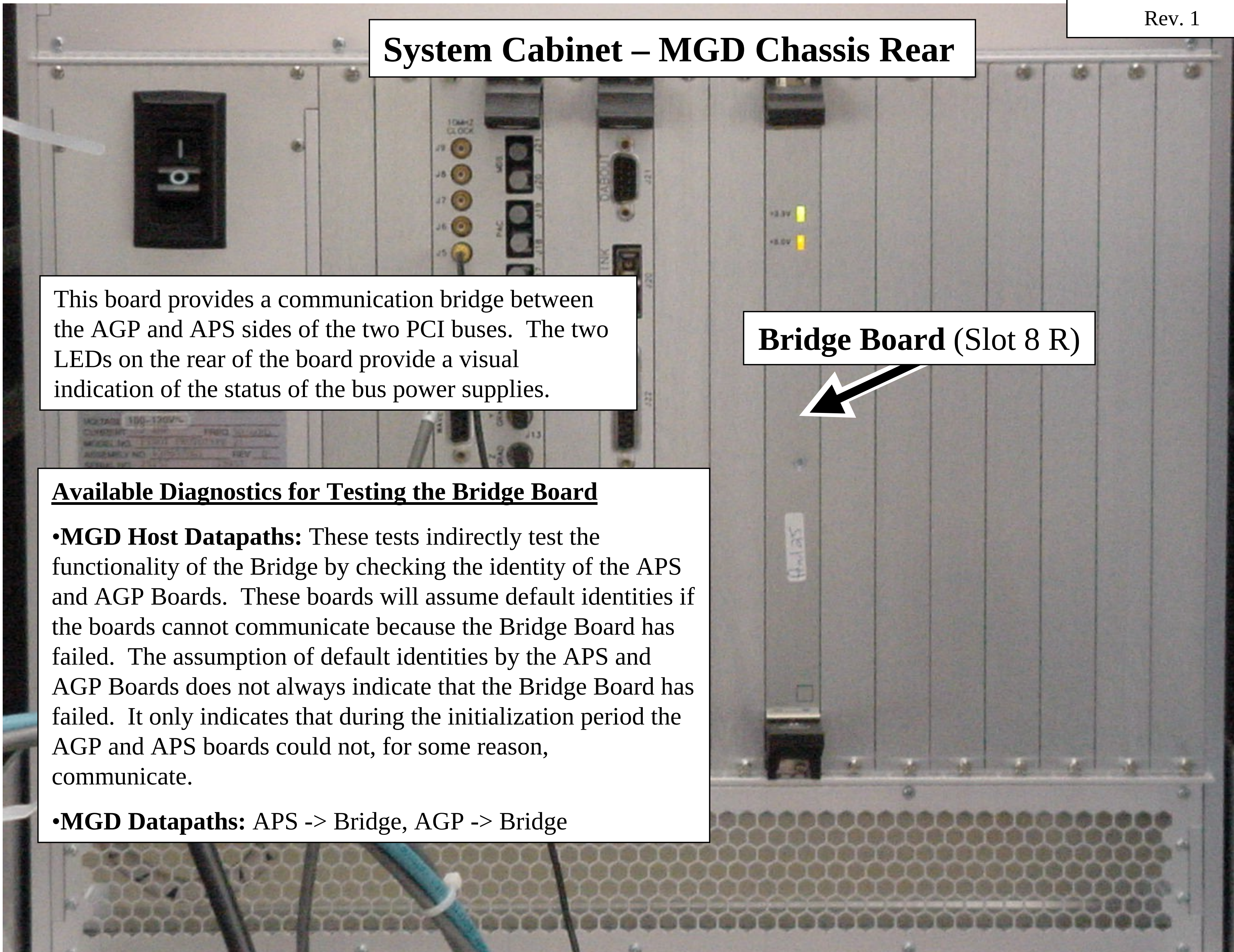
System Cabinet – MGD Chassis Front

Power Supply Modules (PS1 F, PS2 F)



Both modules work in tandem to supply power to the MGD Chassis. The output voltages these boards provide cannot be adjusted. During normal operation the green POWER SUPPLY LED will be illuminated. Some types of power supply failures will cause the red FAULT LED to illuminate. A row of Green indicator LEDs located above the supplies in the MGD Chassis provide a visual indication of each supply voltage. Test points next to each LED provide a convenient measurement location for checking the condition of each supply voltage with a Digital Multimeter. Please note that the MGD hardware does **NOT** support "hot swapping" (i.e. removing and installing power supply with power on) of power supplies. In the event that one of the modules fails, it is recommended to not operate the MGD Chassis with a single supply. One reason for not doing this is that the air flow will be compromised due to the missing supply. This forces the other boards to operate at less than optimal temperatures which could cause damage. Another reason for not doing this is that the remaining power supply is operating over capacity to supply all the MGD Chassis power needs. This can damage the power supply and introduce intermittent failures that may be difficult to isolate. The more options (boards) installed the more stress and possible damage a single supply may encounter.

System Cabinet – MGD Chassis Rear



This board provides a communication bridge between the AGP and APS sides of the two PCI buses. The two LEDs on the rear of the board provide a visual indication of the status of the bus power supplies.

Bridge Board (Slot 8 R)

Available Diagnostics for Testing the Bridge Board

•**MGD Host Datapaths:** These tests indirectly test the functionality of the Bridge by checking the identity of the APS and AGP Boards. These boards will assume default identities if the boards cannot communicate because the Bridge Board has failed. The assumption of default identities by the APS and AGP Boards does not always indicate that the Bridge Board has failed. It only indicates that during the initialization period the AGP and APS boards could not, for some reason, communicate.

•**MGD Datapaths:** APS -> Bridge, AGP -> Bridge

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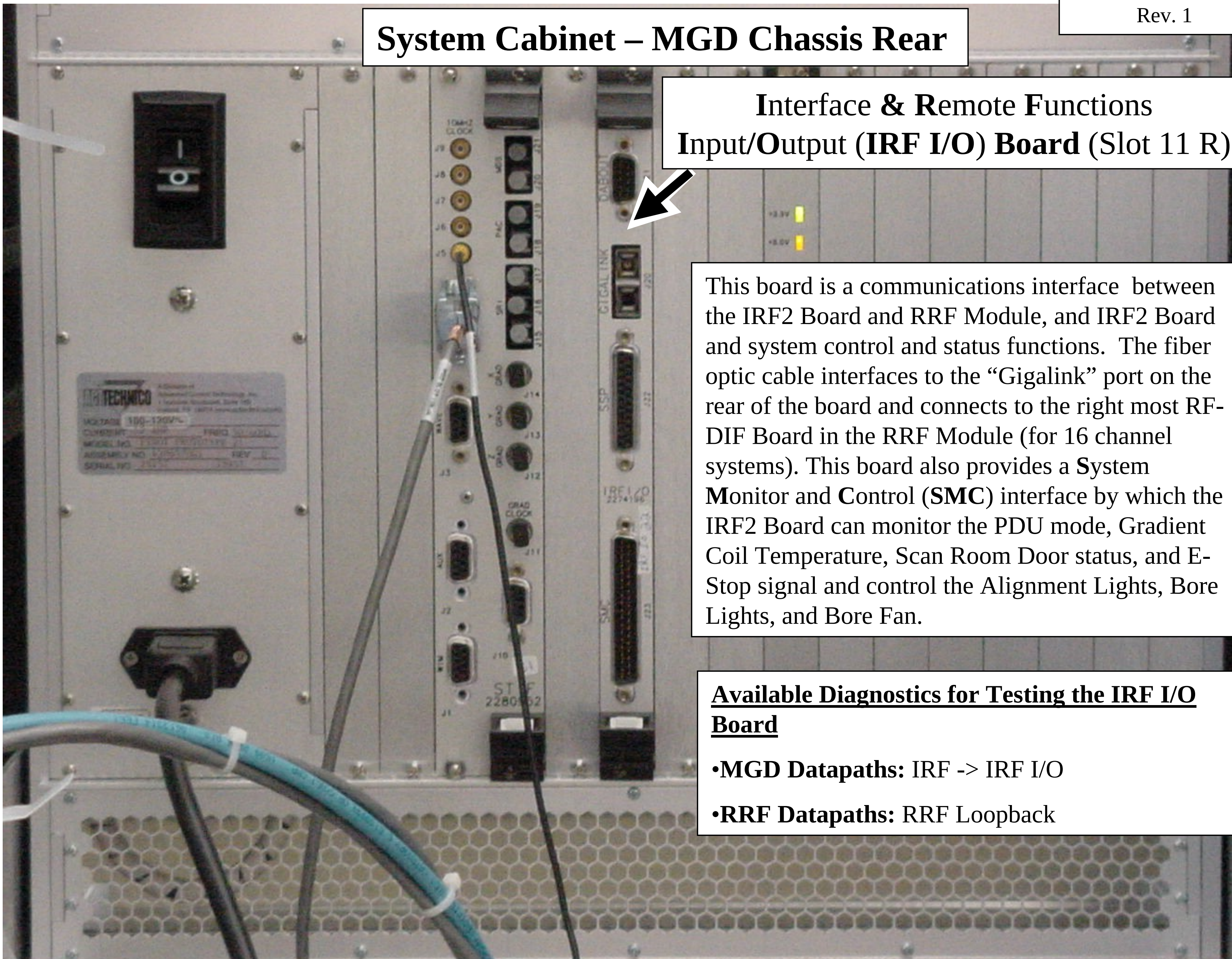
System Cabinet – MGD Chassis Rear



Interface **& R**emote **F**unctions **I**nput/**O**utput (**IRF I/O**) **Board** (Slot 11 R)

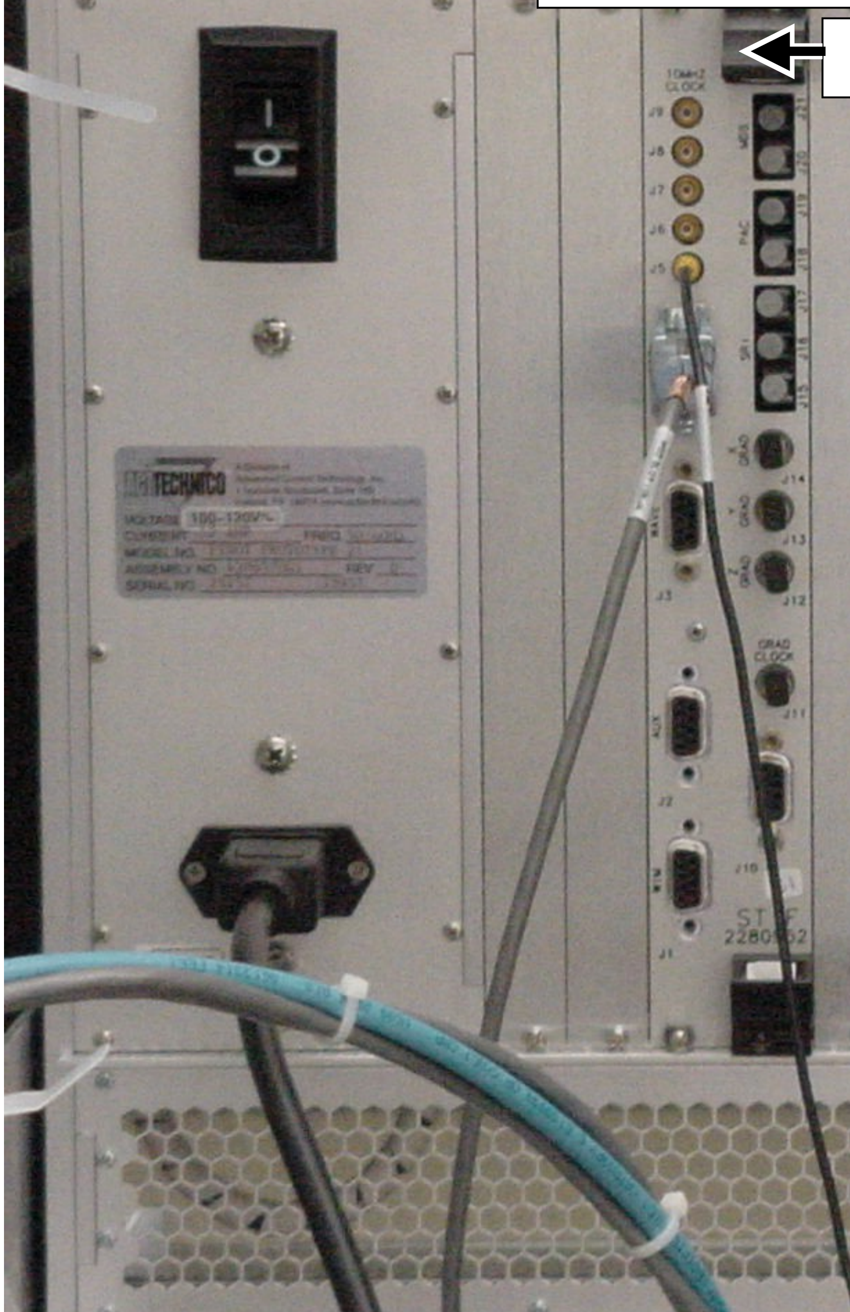
This board is a communications interface between the IRF2 Board and RRF Module, and IRF2 Board and system control and status functions. The fiber optic cable interfaces to the "Gigalink" port on the rear of the board and connects to the right most RF-DIF Board in the RRF Module (for 16 channel systems). This board also provides a **S**ystem **M**onitor and **C**ontrol (**SMC**) interface by which the IRF2 Board can monitor the PDU mode, Gradient Coil Temperature, Scan Room Door status, and E-Stop signal and control the Alignment Lights, Bore Lights, and Bore Fan.

Available Diagnostics for Testing the IRF I/O Board

•**MGD Datapaths:** IRF -> IRF I/O

•**RRF Datapaths:** RRF Loopback

System Cabinet – MGD Chassis Rear



SRF / TRF InterFace (STIF) Board (Slots 13,14 R)

This board provides the fiber optic interfaces (fiber cables are not shown connected) for the fiber cables connecting to the **G**radient **P**rocessor (**GP**) **Board** inside the **HFD Cabinet**, the **SRI** inside the scan room, the **PAC Module** inside the scan room, and the **MDS Link** (upgrades only) that interconnects the System Cabinet to the SRF and HFD Cabinets. The coaxial cable connecting to J5 provides the 60Hz line clock signal from the AC distribution panel inside the cabinet. The gray cable shown in the illustration equipped with the 9-pin sub-D connector provides an RS-232 serial link between the STIF Board and the Host via the Term Server. The function of this link is to supply the signal necessary to start a **TPS Reset** when this is selected from the Operators Workstation. Physiological waveform data, physiological control data, and hardkey control data is transferred from connector J3 (cable not shown connected).

<u>Available Diagnostics for Testing the STIF Board</u>

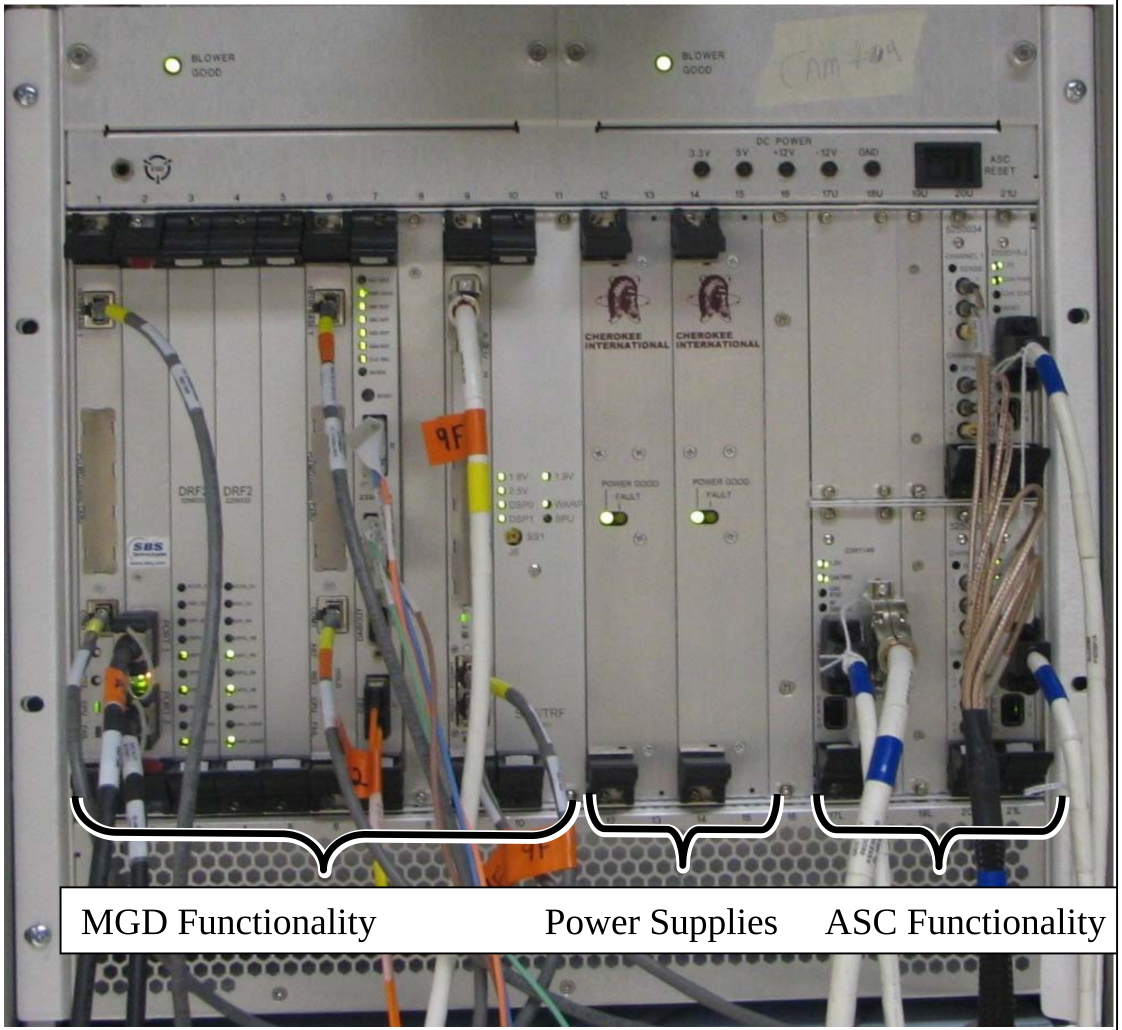
•**SRI Datapaths:** SRI STIF Loopback

•**SRI Functional Tests:** STIF – SRI Fiber Optic

•**Fiber Repeater Diagnostics:** SRI STIF Loopback, SPU-> STIF

•**Fiber Optic Diagnostics:** STIF Fiber Optic Test

System Cabinet – CAM Chassis Differences



Though the functionality didn't change, there are some hardware differences between the MGD chassis and the CAM chassis. These include:

Blower Assembly – Different module with integrated status LED.

Power Supply – Different Vendor with integrated status LED.

Voltage test points and Status LEDs - Test points are part of chassis instead of a separate indicator module. The Chassis Power Supply LEDs were removed as actual PS values can be read from the UPM Diagnostics by selecting Analog Data. The power supplies and blower assemblies include the Status LEDs within their modules.

Bridge Board – Mounted in the front rather than the of rear of the chassis.

Card Slots – New card slot locations for boards plus additional slots added for the ASC functionality.

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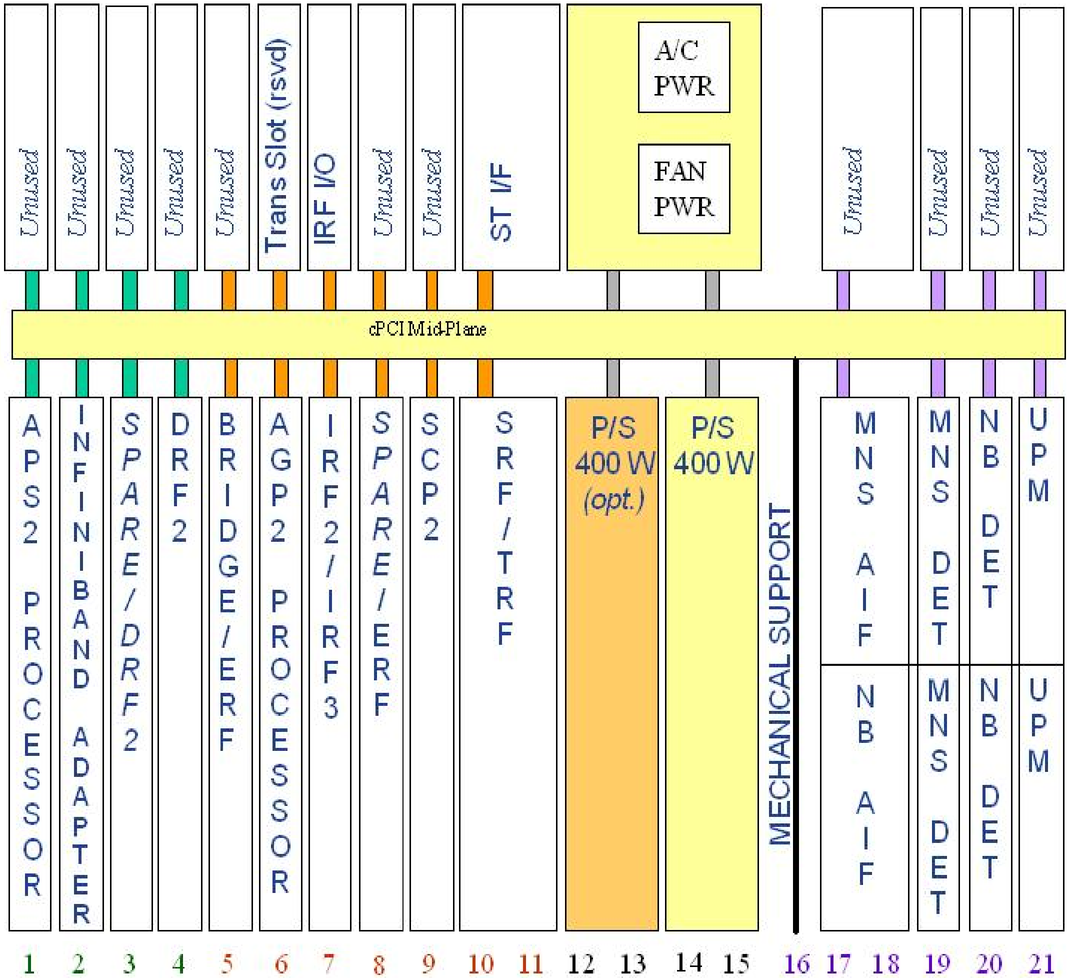
System Cabinet – CAM Chassis Slot Definitions



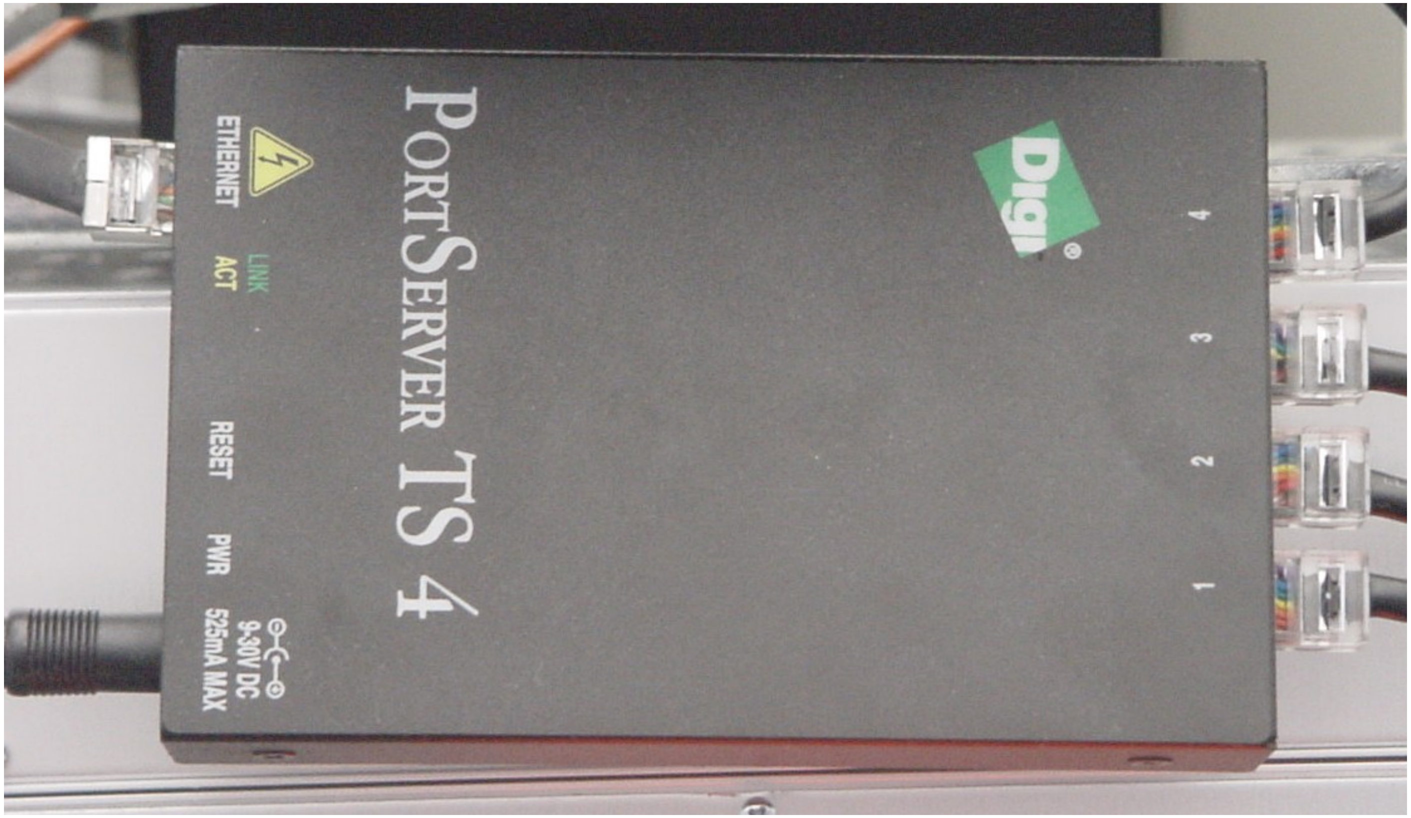
Rear Slots:

Slot 8 : Interface and Remote I/F (IRF I/O)
Slots 10-11: SRF/TRF Interface (STIF)
Slots 12-15: Fan Pwr, AC Pwr, Can Comm

Front Slots:

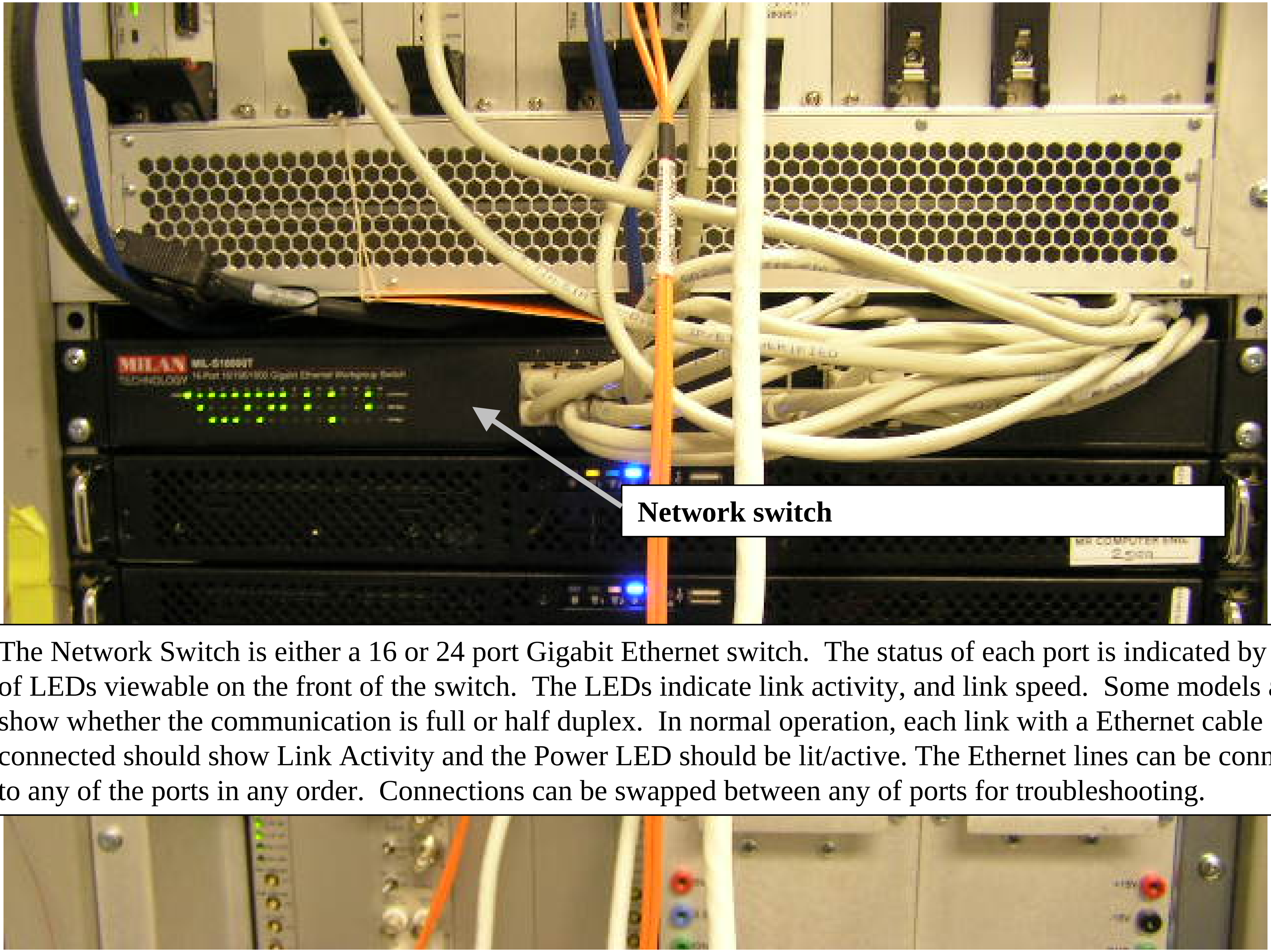
Slot 1: APS2 Processor Board
Slot 2: Infiniband Adapter Board
Slot 3: DRF2 Board or Spare
Slot 4: DRF2 Board
Slot 5: Bridge Board
Slot 6: AGP2 Processor Board
Slot 7: IRF2 Board
Slot 8: Spare
Slot 9: SCP2 Board
Slots 10,11: SRF/TRF Board
Slots 12-13 & 14-15: Power Supply
Slots 17U-18U: Optional MNS Amp I/F Board
Slots 17L-18L: Narrow Band Amp I/F Board
Slot 19U: Optional MNS Detector Board
Slot 19L: Optional MNS Detector Board
Slot 20U: Narrow Band Detector Board
Slot 20L: Narrow Band Detector Board
Slot 21U: UPM Processor Board
Slot 21L: UPM Processor Board

System Cabinet – Term Server



The Term Server permits RS-232C serial connection to each of the processor boards and the STIF Board by utilizing the existing Ethernet link. Serial ports are selected at the Term Server based on the information received in the Ethernet signal. The serial lines on the Term Server are connected as follows: Port 1 connects to the APS, Port 2 connects to the AGP, Port 3 connects to the SCP, and Port 4 connects to the Host serial reset port on the STIF Board. With the exception of the Port 4 connection to the STIF Board, it is not necessary for any of the other serial ports to be functional in order for the system to boot or for the MGD to reset. If the Port 4 serial connection is not functioning, it may not be possible to complete a TPS Reset from the Host. The Ethernet cable from the Term Server can connect to any open port on the Ethernet Switch. The Term Server is physically located next to the Ethernet Switch.

System Cabinet – Network Switch



The Network Switch is either a 16 or 24 port Gigabit Ethernet switch. The status of each port is indicated by a set of LEDs viewable on the front of the switch. The LEDs indicate link activity, and link speed. Some models also show whether the communication is full or half duplex. In normal operation, each link with a Ethernet cable connected should show Link Activity and the Power LED should be lit/active. The Ethernet lines can be connected to any of the ports in any order. Connections can be swapped between any of ports for troubleshooting.

System Cabinet – VRE subsystem